PRATHAMESH SHARAD MESTRY

Electronics & Electrical Product Development Manager

Hardware Architecture · PCB & SI/PI Design · Firmware Engineering · IEC/CE Compliance · EV & Power Electronics · Full Product Lifecycle

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14+	20+	10+	5+
Years Experience	PCB Designs Released	IEC/CE Standards Applied	Products Concept to Market

PROFESSIONAL SUMMARY

Seasoned Electronics & Electrical Product Development Manager with 14+ years of hands-on expertise in the complete product lifecycle — from hardware architecture, multi-layer PCB design, Signal Integrity (SI) & Power Integrity (PI) analysis, analog/mixed-signal circuits, and power electronics, through to embedded firmware engineering, IEC/CE/UL compliance, manufacturing bring-up, and field deployment. Designed and released 20+ production PCBs across EV charging, smart energy metering, power electronics, IoT, and railway safety domains.

Proven team and project leader with experience managing cross-functional R&D teams of 10+ engineers using Agile/Scrum methodologies. Deep compliance engineering expertise with IEC 61851, IEC 62196, IEC 62056, IEC 61000, CE, UL, SAJ, and DLMS/IDIS standards from design stage through certification. Combines core hardware/firmware depth with growing proficiency in AI/ML for edge devices, full-stack IoT software, and data-driven diagnostics. Targeting a senior product engineering or R&D management role in EV, smart energy, power electronics, or industrial IoT.

CORE COMPETENCIES

Hardware Architecture	**PCB Design (SI/PI)**	**Power Electronics**	**Firmware Engineering**	**RTOS & Bare-Metal C**
IEC/CE/UL Compliance	**DLMS/IDIS Standards**	**EV Charging Systems**	**IoT & Cloud Embedded**	**AI/ML on Edge**
Product Lifecycle (PLM)	**Team Leadership (10+)**	**Agile / Scrum / Sprint**	**DFMA / FMEA / RCA**	**Vendor & Stakeholder Mgmt**

HARDWARE ENGINEERING EXPERTISE

PCB Design & Schematic Engineering

- End-to-end multi-layer PCB design (4–8 layers): Altium Designer 23, Cadence OrCAD, KiCAD 7, Fusion 360, EasyEDA, EAGLE, DesignSpark, PADS PCB — consumer, industrial, and high-voltage product classes
- Hierarchical schematic capture, net-class management, differential pair routing, impedance-controlled traces for high-speed signals (USB, CAN-FD, Ethernet, LVDS)
- BOM management: component selection, alternate sourcing, lifecycle management

Signal Integrity (SI) & Power Integrity (PI)

- Signal Integrity analysis: impedance-controlled routing (50Ω single-ended, 100Ω differential), via stub minimization, back-drilling, length matching for high-speed buses (USB 2.0/3.0, SDIO, Ethernet, CAN-FD)
- Pre-layout SI planning: stack-up definition, dielectric selection, controlled impedance trace width calculation using Saturn PCB toolkit and Altium Layer Stack Manager
- Power Integrity analysis: PDN (Power Delivery Network) design, decoupling capacitor strategy, bulk and ceramic capacitor

(NRND/EOL), cost optimization across NPI and mass production phases
- PCB manufacturing file preparation: Gerbers, ODB++, IPC-2581 packages — coordination with PCB fab and PCBA vendors across India and internationally
- Design Rule Check (DRC) / ERC — zero critical violations policy; IPC-2221A compliance for creepage, clearance, and annular ring

placement for minimum impedance across frequency, VRM transient response optimization
- Target impedance calculation for each power rail, PDN impedance profiling, anti-resonance mitigation using multi-value capacitor arrays across supply planes
- EMC-aware layout: ground plane stitching, split plane management, return current path control, guard rings, via fence shielding, filter placement at board entry/exit points
- Post-layout SI/PI validation using MATLAB, LT-Spice, and HyperLynx (awareness); correlation with oscilloscope eye diagrams and power rail ripple measurements during bring-up

Analog & Mixed-Signal Hardware Design

- Precision analog front-end: instrumentation amplifiers, op-amp signal conditioning for CT, Rogowski coil, shunt resistor interfaces for energy metering and power monitoring
- High-accuracy ADC interfacing: 16/24-bit sigma-delta ADCs (ADE7880, ADE9078, CS5463) — input filtering, reference design, low-noise grounding, and calibration hardware
- Power supply design: LDO, isolated/non-isolated SMPS (flyback, buck, boost), PFC stage design for embedded systems and AC-connected products
- Analog protection circuits: TVS, MOV, PTC fuse, transient suppression per IEC 61000-4-2/4/5; over-voltage, over-current, and reverse polarity protection
- Crystal oscillator, PLL clock design, and RTC backup circuitry for precision timing in energy meters and embedded controllers
- Isolated IGBT gate driver design — bootstrap circuits, dead-time control hardware, DESAT protection for power converters and STATCOM applications

Power Electronics & Embedded Hardware

- AC EV Charger hardware (7.4–22 kW): relay/contactor drive, CT current sensing, earth fault detection (RCD/GFCI), CP/PP pilot signal circuits per IEC 61851-1
- SMPS design for embedded rails: isolated flyback for MCU/logic, synchronous buck for high-current rails, EMI filter for conducted emissions compliance
- STATCOM hardware: IGBT H-bridge PCB, isolated gate drivers (ACNV4506, TLP250H), DC-link capacitor bank, snubber design, current/voltage sensing for closed-loop control
- PLC hardware: line coupling transformer, band-pass/band-stop filter networks, HF injection circuits, impedance matching for power-line signal propagation
- MCU selection & architecture: STM32 (F/G/H/L/WB series), ESP32/ESP32-S3, AVR, PIC — peripheral mapping, memory architecture, power budget, and BOM cost analysis
- Sensor interfacing: NTC/PTC thermistors, Hall effect, PT100 RTD, IMU (SPI/I2C), TFT/HMI display hardware (LVGL driver interface), RS-485/CAN transceiver circuits

Hardware Bring-Up, Validation & NPI

- Structured bring-up: power sequencing, rail measurement, oscilloscope probing, current profiling, thermal imaging (FLIR) for hotspot detection
- Board-level debug: JTAG/SWD ICE setup, boundary scan, logic analyzer protocol capture, spectrum analyzer pre-EMC screening

Hardware Tools & Design Environment

- PCB tools: Altium Designer 23 (primary), Cadence OrCAD, KiCAD 7, Fusion 360, EasyEDA, Autodesk EAGLE, DesignSpark, PADS PCB
- Simulation: LT-Spice XVII (power/analog/transient), MATLAB/Simulink (system-level), NI Multisim, Proteus 8 (embedded co-sim), KiCAD Spice

- DFM/DFA: panelization, fiducial placement, test point accessibility, AOI/X-ray considerations — coordinated with CM/EMS partners
- Environmental testing: thermal cycling, vibration, humidity, IP-rating enclosure validation for industrial and outdoor qualification
- First Article Inspection (FAI) coordination; tolerance stack-up analysis, golden sample management, ICT/FCT production test fixture guidance
- Engineering Change Order (ECO) process: schematic/layout change control, Gerber revision management, BOM delta tracking, re-qualification planning
- SI/PI tools: Saturn PCB Toolkit, Altium Layer Stack Manager, LT-Spice for PDN modeling, oscilloscope eye diagram analysis
- Test & measurement: Tektronix/Rigol oscilloscope, Yokogawa power analyzer, LCR meter, FLIR thermal camera, spectrum analyzer, DMM
- Mechanical: Fusion 360 for PCB-to-enclosure fit, 3D STEP import/export, enclosure cutout and mounting boss design coordination
- Documentation: hardware design spec, schematic review checklists, test reports, design history files (DHF) for regulatory submissions

FIRMWARE ENGINEERING EXPERTISE

Firmware Architecture & RTOS

- Bare-metal and RTOS (FreeRTOS, µC/OS-III, Zephyr) firmware on ARM Cortex-M0/M3/M4/M7 (STM32) and Xtensa LX6/LX7 (ESP32)
- Multi-task firmware: priority scheduling, mutex/semaphore, message queuing, watchdog management, low-power sleep modes
- Low-level peripheral driver development: UART, SPI, I2C, CAN/CAN-FD, LIN, RS-485, MODBUS RTU/TCP, ADC/DAC, DMA, Timers
- OTA firmware update pipelines: dual-bank flash, secure boot, firmware signing (RSA/ECDSA), rollback protection on STM32 & ESP32
- Bootloader development, memory-mapped flash management, AES-128/256 encryption, TLS 1.3 for secure cloud connectivity

Embedded Software Quality & DSP

- MISRA-C 2012 compliant firmware — PC-lint, LDRA, Polyspace static analysis; zero critical violations policy
- Unit testing (Unity/CMock), HIL integration testing (OpelRT, MATLAB/Simulink), structured Git branching and regression suites
- Class A & B energy DSP routines on ARM Cortex-M4 DSP: FFT harmonic analysis, RMS, zero-crossing, power factor, THD
- DLMS/IDIS full protocol stack: COSEM object model, HDLC/WRAPPER layers, TOU tariff, tamper detection per IEC 62056
- PLC FSK/OFDM signal processing firmware, IGBT gate drive control, sensor fusion with Kalman filtering
- OCPP 1.6/2.0 protocol stack (WebSocket/JSON-RPC), BLE 5.0 GATT profile, MQTT/TLS IoT cloud firmware

COMPLIANCE & STANDARDS ENGINEERING

EV Charging & EMC Standards

- IEC 61851-1: Mode 1/2/3/4, CP/PP pilot signal design and firmware; IEC 62196-1/2/3: Type 1/2/CCS2 connector compliance
- IEC 61000-3-2/3 emissions, IEC 61000-4-2/4/5/6/11 immunity — PCB design hardening, filter and protection circuit design
- CE marking: DoC preparation, technical file compilation, notified body and NABL lab coordination (SGS, TÜV, BIS, STQC)

Energy Metering & Safety Standards

- IEC 62056 (DLMS/IDIS), IEC 62052-11, IEC 62053-21/22/23 — Class A/B accuracy design, testing, certification
- IPC-2221A (PCB design), IPC-A-610 (assembly), RoHS/WEEE, ISO 9001 process; IEC 62368-1 product safety
- IEC 60664-1 creepage/clearance calculations for HV PCBs; FMEA, FTA, IEC 61508 SIL awareness, ISO 14971 risk management

– SAJ standard homologation, OCPP 1.6/2.0 interoperability testing, IS 17017 (BIS India EV) awareness	– Full type-test management: test plan authoring, lab coordination, defect resolution, certification maintenance

TECHNICAL SKILLS

🟩 Hardware	🟦 Firmware	🔵 Compliance / IEC	🩵 Software / AI	🟣 Management

HARDWARE DESIGN & ELECTRONICS

PCB & Schematic Design	Altium Designer 23, Cadence OrCAD, KiCAD 7, Fusion 360, EasyEDA, Autodesk EAGLE, DesignSpark, PADS PCB
Signal Integrity (SI)	Impedance control, differential pair routing, via stubs, stack-up design, Saturn PCB Toolkit, Altium Layer Stack Mgr
Power Integrity (PI)	PDN design, decoupling strategy, target impedance calc, anti-resonance mitigation, VRM transient optimization, LT-Spice PDN
Circuit Simulation	LT-Spice XVII, MATLAB/Simulink, NI Multisim, Proteus 8, KiCAD Spice — signal, thermal, power analysis
Microcontrollers & SoCs	STM32 (F/G/H/L/WB series), ESP32/ESP32-S3, Arduino, AVR ATmega, PIC, ARM Cortex-M0/M3/M4/M7, DSP processors
Power Electronics	SMPS, AC/DC, PFC, Inverters, IGBT Gate Drive, STATCOM, EV AC Chargers 7.4–22 kW, PLC systems
Bring-Up & Validation	Hardware-in-Loop, DFT, DFM, DFA, DFMA, CE/RE EMC Testing, Type Testing, FLIR Thermal, JTAG/SWD Debug

FIRMWARE & EMBEDDED SOFTWARE

Languages & Standards	Embedded C (MISRA-C 2012), C++17, MicroPython, Assembly (ARM Thumb) — safety-critical firmware
RTOS Platforms	FreeRTOS (v10+), µC/OS-III, Zephyr RTOS, VxWorks — scheduling, IPC, memory protection, power management
IDEs & Toolchains	STM32CubeIDE, STM32CubeMX, ESP-IDF v5, Keil MDK-ARM, MPLAB X, AVR Studio, IAR EWARM, GCC ARM
Communication Stacks	OCPP 1.6/2.0, DLMS/IDIS, MODBUS RTU/TCP, CAN/CAN-FD, BLE 5.0, MQTT/TLS, CoAP, LwM2M, RS-485
Peripherals & Interfaces	UART/SPI/I2C/I2S, ADC/DAC/DMA, RTC, Timers, PWM, USB CDC, Ethernet (LwIP), TFT display drivers (LVGL)
Code Quality & Security	MISRA-C, PC-lint, LDRA, Polyspace, Unity/CMock, secure boot, AES/TLS, dual-bank OTA, firmware signing

COMPLIANCE & STANDARDS

EV & Charging Standards	IEC 61851-1/2/3, IEC 62196-1/2/3, SAE J1772, OCPP 1.6/2.0, SAJ, IS 17017 (BIS India EV)

Energy Metering	IEC 62056 (DLMS/IDIS), IEC 62052-11, IEC 62053-21/22/23, IEC 62059, IDIS Package 2
EMC & Safety	IEC 61000-3-2/3, IEC 61000-4-2/4/5/6/11, IEC 62368-1, IEC 60664-1 — design hardening and type testing
PCB & Quality	IPC-2221A, IPC-A-610, IPC-7711/21, RoHS/WEEE, ISO 9001; FMEA, FTA, IEC 61508 SIL awareness, ISO 14971
Certification Process	CE Marking (DoC, technical file), TÜV/SGS/BIS/STQC lab coordination, type testing, NABL management

SOFTWARE, AI/ML & DATA SCIENCE

Programming Languages	Python 3 (Advanced), Java, C/C++, PHP, JavaScript (ES6+), SQL, Bash
Frontend & Backend	HTML5, CSS3, Bootstrap 5, React.js, Flask, FastAPI, Django REST, Node.js/Express — IoT dashboards & REST APIs
AI / ML & Edge AI	Scikit-learn, TensorFlow 2/Keras, PyTorch, TFLite Micro (STM32/ESP32) — predictive maintenance, anomaly detection
Data Science & Cloud	Pandas, NumPy, Matplotlib, InfluxDB, Jupyter; AWS IoT Core, Azure IoT Hub, Firebase, Node-RED, MQTT, Docker
Model-Based Design	MATLAB R2023b, Simulink, Stateflow, Embedded Coder (auto code gen), Hardware-in-Loop testing

PRODUCT, PROJECT & TEAM MANAGEMENT

Team Management	Cross-functional team leadership (10+ engineers), technical mentoring, performance reviews, hiring & onboarding, OKR setting
Product Management	Product roadmapping, PRD/MRD writing, feature prioritization, GTM strategy, stakeholder alignment, Voice-of-Customer
Project Management	Agile/Scrum, sprint planning & review, Kanban, Waterfall, risk register, SDLC, milestone tracking, MS Project
Tools & Platforms	Jira, Confluence, Trello, JAMA (requirements), GitHub Projects, Slack, Microsoft Teams, Notion
Quality & Process	DFMA, FMEA, RCA, ECO/change control, design review gate process, IPC quality, supplier qualification
Version Control	Git, GitHub, Bitbucket, Tortoise SVN — branching strategy, PR reviews, CI/CD pipeline awareness

PROFESSIONAL EXPERIENCE

Lead Project Engineer
Jul 2022 – Nov 2025

Sai Agro Tele | *Pune, India*

Focus: Hardware Design 100% · Project & Team Management

- **Hardware:** Led end-to-end hardware design of embedded modules and PLC (Power Line Communication) systems — multi-layer PCB layout in Altium Designer, hierarchical schematic capture, BOM management, analog front-end design for HF signal coupling, band-pass/stop filter networks, and line coupling transformer design.

- **SI/PI:** Performed Signal Integrity and Power Integrity analysis on PLC and embedded module PCBs — impedance-controlled routing, PDN decoupling strategy, return current path management, and EMC-aware ground plane design to meet IEC 61000-4 immunity requirements.
- **Hardware:** Executed commissioning, type tests, and hardware bring-up for electronics modules; coordinated with international component manufacturers, suppliers, and certification labs; managed component lifecycle and alternate sourcing to mitigate supply chain risks.
- **Hardware:** Designed analog protection circuits (TVS, MOV, surge), power supply rails (isolated SMPS, LDO), and sensor interfaces; performed thermal analysis and layout optimization; maintained design versioning on GitHub and Altium Vault.
- **Project Mgmt:** Managed full hardware development project lifecycle — milestone planning, sprint reviews as Scrum Lead, stakeholder requirement management in JAMA, documentation in Confluence, and cross-functional coordination with firmware, QA, and procurement teams.
- **Team Mgmt:** Led and mentored a hardware engineering team — assigned tasks, conducted design and schematic peer reviews, tracked deliverables, and maintained team velocity through Agile sprints on Jira.
- **Risk:** Prepared DFMA/FMEA risk analyses for device safety; coordinated vendor qualification, established design review gate checklists, and managed ECO process for hardware revisions.

Product Development Lead

Nov 2021 – Jul 2022

Magenta EV Solutions Pvt. Ltd. | *Mumbai, India*

Focus: Hardware 70% · Firmware 30% · Testing & Compliance · Project & Team Management

- **Hardware:** Owned complete hardware architecture and PCB design of compact AC EV chargers (7.4 kW) — multi-layer board in Altium Designer, schematic capture (relay/contactor drive, CP/PP pilot signal circuits, CT current sensing, RCD/earth fault detection), BOM design, and design reviews for IEC 61851 compliance.
- **SI/PI:** Applied SI/PI best practices on EV charger PCBs — controlled impedance for CAN-FD and USB traces, PDN optimization for STM32 and ESP32 power rails, EMC-hardened layout with guard rings, decoupling capacitor arrays, and IEC 61000-4 design hardening.
- **Hardware:** Managed board assembly, bring-up, and validation — power sequencing verification, oscilloscope probing, FLIR thermal imaging, DFM/DFA review with CM partner; created golden sample and production bring-up checklist.
- **Firmware:** Implemented OCPP 1.6 protocol stack (WebSocket/JSON-RPC) on ESP32 — heartbeat, remote start/stop, smart charging profiles, meter value reporting, BLE 5.0 GATT profile, and OTA firmware update pipeline.
- **IEC/Testing:** Led IEC 61851-1 Mode 3 CP state machine firmware and IEC 62196-2 (Type 2) connector compliance design; coordinated CE marking, SAJ approvals, and IEC type testing — authored technical file and DoC; performed EMC pre-compliance and formal CE/RE testing with external labs.
- **Project Mgmt:** Managed full product development schedule — sprint planning, milestone tracking, stakeholder updates, and cross-functional coordination between hardware, firmware, cloud software, manufacturing, and QA teams using Jira and Confluence.
- **Team Mgmt:** Led 6-member hardware and embedded team — conducted MISRA-C code reviews, schematic peer reviews, sprint retrospectives, and standardized bring-up procedures; reduced prototype-to-production cycle time through process improvements.

Lead Product Development Engineer

Apr 2021 – Nov 2021

goEgo Networks Pvt. Ltd. | *Pune, India*

Focus: Hardware Design 100% · Project & Team Management

- **Hardware:** Led complete hardware development of AC EV chargers — PCB design in Fusion 360, hierarchical schematics (ESP32 MCU, relay drive, CP/PP signal circuitry, CT-based current sensing, earth fault detection), BOM management, IEC 62196 connector compliance, and vendor fabrication support.
- **SI/PI:** Implemented SI-aware PCB layout for ESP32-based charger controller — impedance-matched RF antenna trace, decoupling strategy for 3.3V and 1.2V rails, return current path control and EMC guard rings around switching power supply area.

- **Hardware:** Managed board assembly and full bring-up process — power rail verification, oscilloscope signal capture on CAN/UART/SPI interfaces, thermal imaging, and EMC hardening iterations for CE pre-compliance; coordinated PCBA with CM partners across 3 hardware revisions.
- **Hardware:** Designed analog front-end circuits: shunt/CT current measurement, pilot signal detection comparator circuit, relay/contactor drive with snubber, LED status driver; validated all circuits against IEC 61851-1 requirements with lab instruments.
- **Project Mgmt:** Managed hardware development project from concept through CE pre-compliance — defined weekly milestones, maintained BOM and schematic revision control, coordinated with firmware and cloud teams on interface specifications, and managed vendor timelines.
- **Team Mgmt:** Led hardware engineering team — assigned design tasks, reviewed schematics and layouts, conducted design review gates, and maintained cross-functional alignment between hardware, embedded software, and operations teams.

Senior Embedded Engineer – Hardware R&D

Jan 2011 – Mar 2021

EMBEDD Solutions Pvt. Ltd. | *Mumbai, India*

Focus: Hardware 70% · Firmware 30% · Testing & Certification · Team Leadership

- **Hardware:** Designed DLMS/IDIS-compliant smart energy meters and STATCOM reactive power correctors — multi-layer PCBs in Altium/OrCAD, precision analog front-end (CT/Rogowski coil interface, 24-bit sigma-delta ADC — ADE7880/CS5463), isolated power supplies, RTC backup, anti-tamper hardware (magnetic sensor, neutral missing detection), and surge/ESD protection per IEC 61000-4.
- **SI/PI:** Applied SI and PI rigor on energy meter PCBs — separated analog and digital ground planes with controlled stitching, minimized return current loops on metrology ADC traces, designed multi-value decoupling networks for metrology IC power rails, optimized PDN to meet ADE7880 datasheet noise requirements, and performed post-layout LT-Spice PDN impedance verification.
- **Hardware:** Designed STATCOM hardware — IGBT H-bridge PCB with isolated gate drivers (TLP250H), DC-link capacitor bank, snubber circuits, gate drive bootstrap supply, high-voltage creepage/clearance per IEC 60664; designed PLC line coupling transformer and HF filter networks for Power Line Communication modules.
- **Hardware:** Managed complete BOM lifecycle across 10 years — component selection, alternate part qualification, NRND/EOL mitigation, cost reduction reviews; supported CM/PCB fabrication vendors with DFM feedback; coordinated FAI and production test fixture (ICT/FCT) design for each product.
- **Firmware:** Developed Class A & B energy measurement firmware libraries on STM32 ARM Cortex-M4 — MISRA-C DSP routines for active/reactive/apparent power, RMS, THD, power factor meeting IEC 62052/62053; implemented full DLMS/IDIS protocol stack (COSEM, HDLC/WRAPPER, TOU tariff, tamper detection per IEC 62056).
- **Testing:** Obtained and maintained IEC 62052-11 and IEC 62053-21/22 certifications — authored accuracy class test plans, managed calibration infrastructure, performed CE/RE EMC testing, and coordinated NABL/BIS lab submissions for full type approval across multiple product variants.
- **Team Mgmt:** Led 10-member hardware R&D team over 10 years — sprint planning, schematic/layout design reviews, MISRA-C code reviews, DFMA/FMEA sessions, mentoring junior engineers, and establishing Git version control and structured design review gates within the organization.

KEY PROJECTS & ACHIEVEMENTS

PLC Smart Home Lighting Control System

Jul 2022 – Mar 2023

End-to-end PLC hardware design and embedded firmware for residential lighting control. HF carrier modulation onto 230V power lines with SI-aware analog front-end.

– Hardware (100%): Line coupling transformer design, band-pass/stop filter networks, analog front-end PCB with SI-optimized layout, EMC hardening per IEC 61000-4-4/5
– Project Mgmt: Led hardware design and validation milestones; MATLAB Simulink signal integrity modelling pre-prototype; coordinated PCB fab and bring-up

AC EV Point Charger – 7.4 kW / 22 kW (CE/IEC Certified)

Apr 2021 – Nov 2021

Full lifecycle development of AC EV charger: hardware architecture through IEC/CE-certified commercial launch.

- Hardware (70%): STM32+ESP32 dual-controller PCB in Altium, SI/PI-optimized layout, IEC 61851-1 CP/PP circuits, CT sensing, RCD hardware, IEC 60664 HV clearances, EMC hardening
- Firmware (30%): IEC 61851-1 Mode 3 state machine, OCPP 1.6 over WebSocket/TLS, BLE 5.0 GATT, FreeRTOS, OTA firmware update
- Compliance & Testing: CE marking, IEC 62196-2 type testing, IEC 61000-4 EMC, SAJ homologation — full technical file and DoC

Indian Railway RDSO Smart Signalling (IoT Upgrade) — Jan 2019 – Apr 2021

Hardware design and firmware for wireless IoT signalling nodes replacing legacy wired railway infrastructure per RDSO specifications.

- Hardware (70%): RF wireless node PCB, antenna matching, ESD/surge protection for outdoor railway environment, fail-safe relay circuits
- Firmware (30%): Fail-safe state machine with redundant communication paths, watchdog-protected safety transitions, RDSO-compliant message protocol

STATCOMS – Reactive Power Generator & PFC System — Jun 2017 – Jan 2019

Power electronics hardware and real-time DSP firmware for STATCOM-based reactive power compensation.

- Hardware (70%): IGBT H-bridge PCB, isolated gate drivers, DC-link capacitor bank, snubber circuits, SI/PI-compliant layout, IEC 61000-3-2 emissions-compliant power stage design
- Firmware (30%): Real-time FFT reactive power measurement, IGBT dead-time control, fault-safe state transitions on STM32 Cortex-M4 DSP

BLE-Enabled DLMS/IDIS Smart Energy Meter (IEC Certified) — Jan 2014 – Jun 2017

IEC-certified Class A & B smart energy meter with IoT connectivity and cloud monitoring dashboard.

- Hardware (70%): Precision 24-bit ADC metrology PCB (ADE7880), SI/PI-optimized analog layout with split ground planes, BLE antenna, surge/ESD protection, anti-tamper hardware
- Firmware (30%): DLMS/IDIS full stack, Class A/B DSP energy library (IEC 62052/62053), BLE GATT, MQTT cloud push, tamper detection
- Compliance & Testing: IEC 62053-21/22 accuracy certification, IEC 61000-4 immunity, NABL lab coordination, DLMS conformance testing

EDUCATION

M.E. – Electronics & Telecommunication Engineering (EXTC) — **2011 – 2014**
Vidyalankar Institute of Technology, Mumbai
71% · First Class with Distinction

B.E. – Electronics & Telecommunication Engineering (EXTC) — **2008 – 2011**
Vidyalankar Institute of Technology, Mumbai
75.54% · First Class with Distinction

Diploma – Electronics & Telecommunication Engineering — **2004 – 2008**
Larsen & Toubro Institute of Technology, Mumbai
88.54% · First Class with Distinction

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Embedded Systems	Embedded C & C++	PCB Design & Testing	Altium Designer

	(MISRA)		Advanced
Signal & Power Integrity	IEC 61851 EV Charging	DLMS/IDIS Compliance	EMC Design & Testing
MATLAB Onramp	MATLAB Simulink	TensorFlow / Keras ML	Python for Data Science
AI on Edge (TFLite)	AWS IoT Core	Agile & Scrum Master	Science of Leadership

LANGUAGES

English (Fluent – US & UK) · Hindi (Fluent) · Marathi (Native) · Kannada (Basic) · Gujarati (Basic)

DECLARATION

I hereby declare that all the information provided in this resume is accurate and true to the best of my knowledge and belief.

Prathamesh Sharad Mestry · Maharashtra, India